BORDER OPTICS — Research Paper

Satellite Verification of India’s Vibrant Villages Programme: A Compositing-Window Robustness Assessment of Border-Village Development

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Abstract

Ask the Government of India if things have worked out for them with their flagship scheme for the Border Villages, the honest answer (on parliamentary record) is No, it has never been verified anywhere. The Vibrant Villages Programme (VVP-I) has approved around ₹4,800 crore in support of 2,967 villages in five Himalayan border states/union territories including 662 villages marked ‘priority’ for Phase-I of the programme since 2022–23. There is no impact assessment in any manner. This study utilizes two satellite proxies: Sentinel-2 built-up-index (NDBI) and VIIRS night-lights at 258 individually geocoded villages across three core states (Arunachal Pradesh, Sikkim, Uttarakhand), an illustrative sample of 7 villages in Himachal Pradesh, and compares them with a district-restricted non-VVP control group of 735 villages across the same 14 districts. Everything was calculated twice, with two defensible methods of compositing the same 2021-2025 satellite record, specifically to check whether the answer depends on that choice. On the headline test, it mostly doesn’t, once both windows are extracted completely and on the same day as the control group: neither shows a significant treated-only built-up-area increase (full-year Wilcoxon $p = 1.000$; summer-matched, June–September, $p = 0.435831$, $n = 251$). That second number is the one that changed. An earlier version of this same summer-window test, run before all 31 Sikkim villages’ monsoon-season imagery had backfilled into the Sentinel-2 archive, returned $p < 0.000001$ on a 154-village subsample missing Sikkim entirely — a result that looked like a clean, surviving signal until the complete sample was actually available (see Development Log Entries 21–22 for the full account, including two extraction-script bugs found and fixed in the process of getting there). Against the control group, the story is the same: neither window’s NDBI gap holds (full-year did coefficient = +0.0088, $p = 0.275$; summer = -0.0038, $p = 0.447$, now on the complete 251-treated/735-control sample), nor does the buffer-radius sweep (250m/500m/1km, all null once every radius uses the same complete sample: $p = 0.126/0.436/0.899$), nor the leave-one-district-out check (0 of 14 reruns significant, against 14 of 14 on the earlier, incomplete sample), nor a randomization-inference test that doesn’t depend on cluster asymptotics ($p = 0.115$). Night-lights tells almost the same story on its own treated-only terms — no absolute change in either window (Wilcoxon $p = 0.050$ full-year, $p = 0.9999$ summer) — with one exception: a positive treated-vs-control gap survives in the full-year window under district fixed effects (+0.1653, $p = 0.0364$), weaker under a leave-one-district-out check (significant in 10 of 14 reruns, coefficient range +0.116 to +0.217) and a randomization-inference check ($p = 0.0155$) than the old summer NDBI result ever needed to be, absent under the no-fixed-effects HC3 specification ($p = 0.1033$), and sitting on the single most imbalanced treated-vs-control

2021 baseline of any test in this study ($p < 0.00001$) — a real but fragile candidate, not a confirmed one. The summer-window night-lights gap, previously significant against the control group ($p = 0.0044$), is null on the complete data ($p = 0.366$). One new result appears in the same complete summer data: built-up change now correlates with distance to the border/LAC (H3, $\rho = 0.291$, $p = 0.0000026$) — the opposite of both H3's own prediction and this test's own earlier result on the incomplete sample ($\rho = 0.038$, $p = 0.641$, previously described as "a truly stable null"). It has not been stress-tested the way the DiD result was, and is reported here as a fresh, open finding, not a confirmed one. Budget still doesn't track outcome in a way worth building a claim on: with all three core states now having valid summer data (previously two, Sikkim missing), Arunachal Pradesh's roughly tenfold larger sanctioned budget over Uttarakhand's corresponds to a smaller mean NDBI change (+0.0023 vs +0.0125), and Sikkim's smallest budget of the three corresponds to a negative one (-0.0347) — descriptive only, $n = 3$, and the correlation's own sign flips between the two compositing windows, which is itself more evidence that this comparison isn't resolvable at this sample size than evidence of any real pattern. None of this is the verdict this study originally reported, and that reversal is itself the more important finding: once the extraction is complete and treated/control data are pulled the same day, no satellite-detectable effect of VVP-I survives being checked two ways, across two proxies, against a control group, and at three buffer radii — which is still a real, reportable answer to a question Parliament's own record says was never asked.

Keywords: Vibrant Villages Programme, border development, securitization theory, satellite verification, NDBI, VIIRS night-lights, remote sensing, robustness testing, difference-in-differences.

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1. Introduction

India's border villages have been part of the discourse of policy for long, as liabilities – they are found in remote areas, are sparsely populated, and prone to being lost out – towards a 'contested frontier'. That's the solution the Vibrant Villages Programme (VVP-I) has proposed and taken to the Union Cabinet for approval, which cleared the project in February 2023, turning 2,967 villages in Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh and Ladakh into development priorities, not security afterthoughts. One thing is missing from that story, however, more than two years after sanction gone by, there has been no independent basis of determining whether the promised development actually occurred. This time, Parliament directly asked, has VVP made an impact? The reply was unambiguous: No impact assessment has been done (Lok Sabha Unstarred Question No. 508 of 3 February 2026, Ministry of Home Affairs).

This study is just that. Test—village by village—whether any physical development appears on the ground because of sanction; whether the size of a village correlates with any of the state's budgets; whether its location is more closely related to being near the border as opposed to the level of development required. Here is the place of securitization theory: If border infrastructure is a geopolitical sign, rather than a welfare policy measure, then results should be as correctly predicted as crucially as need is.

There are five research questions that emerge from this, namely, does expansion of built-up area happen at priority villages since sanction (RQ1); does the size of this expansion correspond to the budget allowed (RQ2); do the night-lights confirm the picture of built-up area changes (RQ3); is development concentrated along the border and not where it is most needed (RQ4); and is all of this growth linked to VVP-I or is it linked at the village level generally (RQ5)? Four hypotheses provide operationalizations of the questions. Time since the implementation of the policy makes a big difference on the measure of built-up area. This increase will not grow proportionally to budget: the increase will be relatively smaller in the states with relatively bigger budget. Villages closer to the border/LAC seem to transform more rapidly than those further away within the same State, as securitization theory would suggest. The test that identifies a programme effect beyond the general trend – or difference (H4) – predicts that change will be significantly higher in VVP-I priority villages as compared to a district-restricted set of non-VVP villages within the same districts and the same period.

2. Literature Review

2.1 Securitization Theory and Border Development

By shifting the agenda from “ordinary” politics to securitisation, opportunities to unlock resources and “extraordinary” measures that wouldn’t be politically possible arise (Buzan, Wæver, & de Wilde, 1998). Spending on border development is an almost perfect fit into this picture: As spending on welfare, it is legitimate; as spending to build strategic links, it is also legitimate; and money is seldom separated in policy literature. Here, VVP-I is used as a test case of that theory: is geographic proximity to the border an even better indicator of measurable development than the conventional indicators of development — existing infrastructure that would be in place anywhere, for example —?

2.2 The Vibrant Villages Programme: Policy Context and the Accountability Gap

However, the paper trail in VVP-I’s scope lies not altogether in one place in the parliamentary replies. The complete village-wise annexure has been provided in the Rajya Sabha Hansard (Question No. 2321, Ministry of Home Affairs dated 9 August 2023). The 75 priority villages of Himachal Pradesh have been confirmed in Lok Sabha Question No. 2104 (2023) and in Rajya Sabha Question No. 401 (2025) but there is no list of the villages at the name level. Ladakh’s 35 sanctioned villages are officially confirmed as such in Lok Sabha Question No. 4360 (2025), but without measures published giving the names of the villages. And answering the question that comes to everyone’s mind when reading this article — Lok Sabha Unstarred Question No. 508 (3 Feb 2026) — Shri Baijayant Panda asked whether VVP’s effect has been assessed and the Ministry’s reply was straightforward — “No impact assessment has been done”. As in searching any of the governmental sources, there are none that present an independent, satellite-verified account of actual advances made over the predetermined scope. This study comes to fill that void.

2.3 Remote Sensing Approaches to Built-Up Area and Economic Activity Detection

Here two indices take the analytical weight, selected for various reasons. A widely-used proxy for built-up surface extent in multi-temporal comparison is the Normalized Difference Built-up Index derived from the reflectance in the short-wave infrared and near-infrared bands (Zha, Gao, & Ni, 2003). The source of radiance as measured by the VIIRS Day/Night Band is somewhat related, but quite different: the amount of electrification and economic activity, and it has excellent dynamic range and resolution compared to the older DMSP-OLS (Elvidge, Baugh, Zhizhin, Hsu, & Ghosh, 2017). This study uses two indices that work well with each other (NDBI versus vegetation phenology and snow; VIIRS versus cloud cover and sensor saturation at low radiance). If either one disagrees with the other, it doesn't necessarily mean either is wrong — but where they agree, that carries more weight than either alone.

2.4 Compositing-Window Sensitivity in Multi-Temporal Satellite Analysis

Change detection for spectral data is difficult for high-relief terrain in certain ways: First, snow cover and cloud frequency vary both by season and elevation; and second, a choice of compositing that isn't elevation-sensitive can create or obscure the signal of true change. This study doesn't consider that sensitivity as an averageable noise. It's being tested directly, as a characteristic of the result itself, and also, which is more important, one that may be reversed between two enthusiastically defendable compositing windows — if either of them were a sole source of confirmation, then that would be the test, wouldn't you focus it on them?

3. Data and Methodology

3.1 Study Design

Only three States, namely Arunachal Pradesh, Sikkim, and Uttarakhand involving 251 villages alone which can be identified on the basis of the village-wise information officially available, are part of the core statistical sample. The 7 confirmed villages of Himachal Pradesh are summarized separately as an exemplar village study, rather than included as part of the core sample, and Ladakh is not included in any village-level study (Section 6.1) because of a documented data-availability gap. The difference between the villages in each group before and after treatment is compared against a difference-in-differences specification (H4) that includes district fixed effects — removing each district's own time-invariant baseline level, not a district-specific *trend*, since the model's post-period term is shared across every district and village alike — and a comparison to a district-restricted non-VVP control group of 735 villages in the 14 districts, to net out whatever common secular trend the wider region experienced over the same period, VVP-affiliated or not. Two additional checks take the translation of "2021-vs-2025" up a level: A third, independent time point (2023) follows; the extraction buffer is fixed at 500m as throughout, and is tested out again at 250m and 1km to determine if the change of the extraction buffer's magnitude changes the answer as well as the compositing window.

3.2 Data Sources

Variable	Source	Temporal Coverage
Village identification	State VVP-I portals; Rajya Sabha/Lok Sabha Q&A annexures	2023–2025
Village coordinates	OpenStreetMap Nominatim; ISRO Bhuvan Village Geocoding API	Current
Non-VVP control village identification	OpenStreetMap Overpass API, district-matched to treated sample	Current

Built-up index (NDBI)	Sentinel-2 SR Harmonized	2021, 2023, 2025
Night-lights radiance	VIIRS DNB monthly composites	2021, 2023, 2025
Border/LAC geometry	Natural Earth 10m Admin-0 Boundary Lines	Current
Budget/project counts	State-wise VVP-I sanction figures (parliamentary record)	2023

The village-level datasets first had to be compiled and geocoded.

3.3 Village-Level Dataset Compilation and Geocoding

Although the primary geocoding was done via Nominatim, Bhuvan was used as a Census-linked fallback — found necessary because Bhuvan doesn't filter by state (a test query for the Arunachal Pradesh village 'Kharman' initially returned an unrelated same-named village in Haryana), so manual district validation was applied throughout. All the results, both from sources, were manually validated in the districts before accepting and any village list was cross validated with the official aggregate figures first. Of the 559 villages attempted across the four states, 258 were successfully geocoded — 186 of 455 in Arunachal Pradesh, 31 of 46 in Sikkim, 34 of 51 in Uttarakhand, and 7 of 7 in Himachal Pradesh.

3.4 Satellite Change Detection

A buffer of 500m around each geocoded village was applied to collect mean NDBI and VIIRS night-lights radiance by using Google Earth Engine 2021 compared to 2025. The NDBI is derived from Sentinel-2 Level-2A Surface Reflectance bands B11 (SWIR1) and B8 (NIR), where cloud was masked using the QA60 band. Night-lights were derived from VIIRS Day/Night Band (NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG).

3.5 Compositing-Window Robustness Testing

Both of these measures were calculated twice – without and with the subtraction of an annual season matched measure (full composite, season matched), designed specifically to remove real change from seasonal artifact (snow in full composite, monsoon cloud in season matched composite). If there was no valid composite in a window, this village did not get zeroed out, it would be marked null.

3.6 Border-Proximity Analysis

The distance calculations were done in GeoPandas and the distance to only those segments of Natural Earth Admin-0 on Indian territory that are relevant to the village was computed. However, there is one proviso that applies to all the numbers that follow this: The line of actual control (LAC) between India and China here is contested, and there is no hard determinate agreed border line. Thus, all of the distances discussed in this study are only relative and comparative and in no way an authoritative delineation of the location of the border.

3.7 Non-VVP Control Group and Difference-in-Differences Design

A treated-only comparison can't rule out a clear alternative explanation: even in the absence of any real VVP-I effect, the same shift could have been caused by conditions shared by all these villages, irrespective of any treatment — anything from unrelated infrastructure spending, unrelated state initiatives or chance regrowth after a low year would do the same. These villages have been dynamically assembled from the same 14 districts as the sample villages using the OpenStreetMap Overpass API, excluding any village that is on a VVP-I priority list, and then processed through the same set of nodes, 500m buffer, NDBI/VIIRS definitions and before/after windows. Results were reshape to be a panel of two periods with a treatment indicator and then tested as

ndbi (or lights) ~ treatment + post + treatment×post + district fixed effects

Using the same district level variable (14 clusters), standard errors were clustered. But the coefficient of interest is that of the treatment × post, which is the difference between the treated group's change and the control group's change in the same time frame, net of the district fixed effects component of any baseline growth trend that varies across districts. A parallel HC3 heteroskedasticity-robust specification without fixed effects is reported together with that as 14 clusters is below the 30–40+ recommended for cluster-robust asymptotics to be fully trustworthy. A fully parallel-pre-trends placebo (the sort of design the project's own earlier control-group research have run elsewhere using a multi-period pre-treatment panel) can not be run on this design. Here the satellite extraction in the control group only includes the same before/after pair that the treated sample has, thereby the level-balance check (Mann-Whitney U) is a weaker at best substitute for a baseline check (2021).

3.8 Multi-Year Trend Extraction

Two single years, 2021 and 2025, are clearly an open possibility how to construct or destroy an apparent change that actually has nothing to do with real development, but can be triggered by weather variation. A third time point was selected, namely 2023, about halfway between the two, extracted for the same 251-village core sample under both compositing windows, and yields one value per village, annually, not another single 'before' and 'after' delta. A per-village linear trend was originally fitted over the three years and the signed ranks Wilcoxon test was used to test the slope against zero, as is often done with this kind of time series model. An additional and alternative analysis was a village-fixed-effects panel regression (value ~ year, clustered SE by village) applied to the slopes of the linear trend model. The two year spans 2021-2023 and 2023-2025 were also tested independently to generate a maximum concentration with a change in one half of the window.

3.9 Buffer-Radius Robustness Testing

In the other tests in this study, the 500m buffer was used as a fixed methodological choice throughout, not tested against other buffers until now. The same extraction for the summer window was repeated at 250m and 1km for the same core sample, and a Wilcoxon signed-rank test was performed on each radius. A complication arose because the 250m and 1km extractions were run at a later date than the original 500m extraction — against the identical fixed 2021/2025 date ranges, but a Sentinel-2 archive that keeps backfilling scenes close to the present — so a raw village-count comparison across the three buffer radii is confounded by that archive-timing difference rather than isolating buffer radius alone. It will come down to making sure that the three radii are restricted to the subset of villages that have data for all three, fixing sample composition and leaving only buffer radius as it varies.

3.10 Statistical Testing

Prior to/after change, Wilcoxon signed-rank test of difference (Wilcoxon, 1945) – paired, non-parametric, appropriate because the sample was not normally distributed. Due to their robustness to non-linear monotonic relationships, Spearman's rank correlation (Spearman, 1904) was used for budget correlation (RQ2, descriptive only – only two States had sufficient valid data) and border-proximity correlation (H3). Control-group comparison (H4) specified with Difference-in-Differences OLS from Section 3.7 with cluster-robust / heteroskedasticity-robust standard errors in parentheses.

4. Results

4.1 Village Coverage and Sample Composition

Overall 258 of 559 villages were successfully located using the geocoding pipeline. Arunachal Pradesh was home to very few civilian entries, and the fact that nothing can possibly be verified without any concrete lists of inhabited points on the border except for those administered by the BRTF camps, army staging huts, and labour camps indicates that they aren't in any civilian geospatial database anyway. This is not really a gap, more of a direct representation of what "village-level development" looks like on a securitized border.

4.2 Built-Up Area Change: No Significant Change in Either Window, Once the Data Is Complete

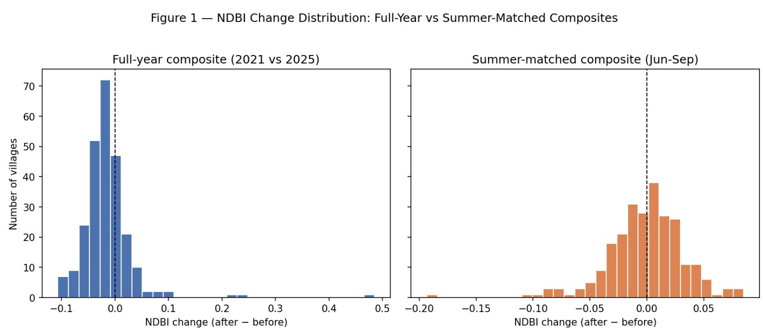

Figure 1

Figure 1. Distribution of NDBI change across all geocoded villages, full-year and summer-matched composites.

Two compositing windows, and now the same conclusion. Full-year: no significant change in the 251-village core sample (Wilcoxon signed-rank, $p = 1.000$), trending slightly negative at the median — this number has never moved across any version of this study. Summer-matched (June–September): also no significant change, once the sample is complete ($n = 251$, $p = 0.435831$). That second number is the one that changed, and the reason it changed is worth stating plainly rather than glossing over. An earlier version of this test, run before this paper's Development Log Entry 22, reported $p < 0.000001$ — but only on a 154-village subsample, because at that point in time all 31 geocoded Sikkim villages returned zero cloud-free Sentinel-2 imagery for the summer window and dropped out entirely. That "zero" turned out not to be a fixed fact about the data: it was a snapshot of how far the Sentinel-2 archive had backfilled scenes for those dates on the day the extraction was run. A later, complete re-extraction —

same collection, same cloud mask, same 500m buffer, same date ranges, pulled the same day as the control group's own summer data — returned real, non-null NDBI values for all 31 Sikkim villages, whose mean summer NDBI change (-0.0347) is the most negative of the three core states (Figure 2) and is what pulls the pooled result from highly significant to a clean null. Full details, including two extraction-script bugs found and fixed while getting to this complete sample, are in Development Log Entries 21 and 22; the 154-village result and everything built on it in earlier drafts of this paper is superseded, not merely refined, by this section's current numbers.

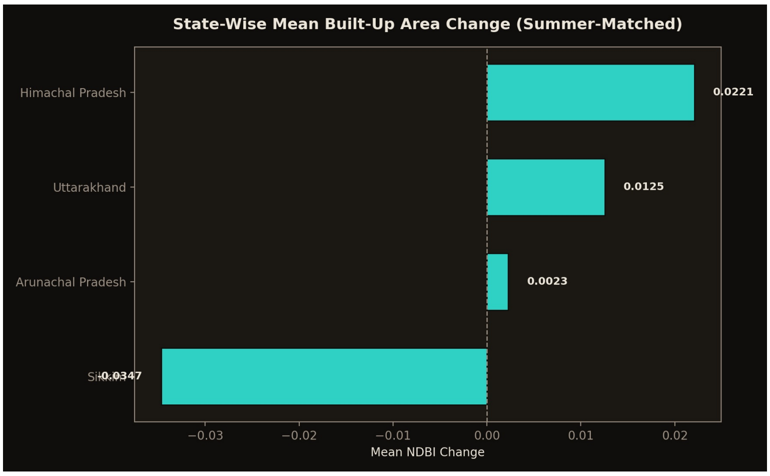

Figure 2

Figure 2. State-wise mean built-up-area change (summer-matched composite), by state.

4.3 Night-Lights Change: A Stable Null

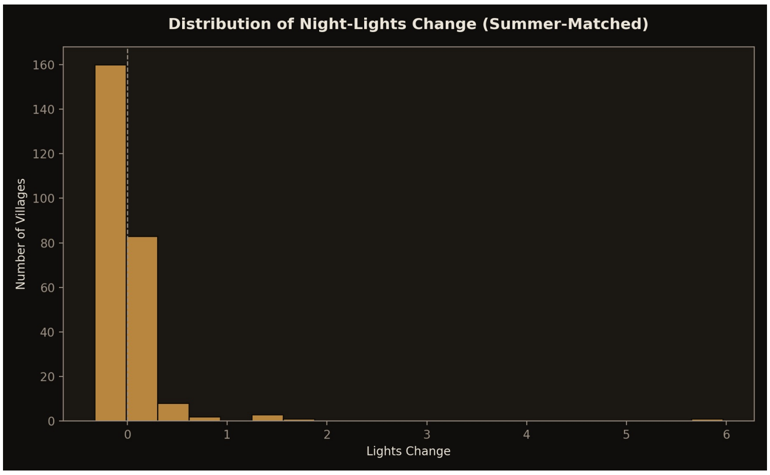

Figure 3

Figure 3. Distribution of VIIRS night-lights change (summer-matched composite).

Night-lights were maintained at the same level in the absence of NDBI: no significant change in this window (full-year p = 0.050 — borderline; summer-matched p = 0.9999 — a clean null). But like many things, it isn't affected by the phenology of vegetation and snow, and that consistency is why it has been viewed as the more reliable of the two proxies — as opposed to a vegetation/snow built-up index that is only as accurate as its component parts. The full-year borderline case (p = 0.050) was also checked against a log1p-transformed version of the same VIIRS radiance values, since radiance is right-skewed — it stays non-significant on that scale too (p = 0.068), so the log transform doesn't flip this conclusion either way; see Section 4.6 for the same check applied to the control-group DiD version of this test.

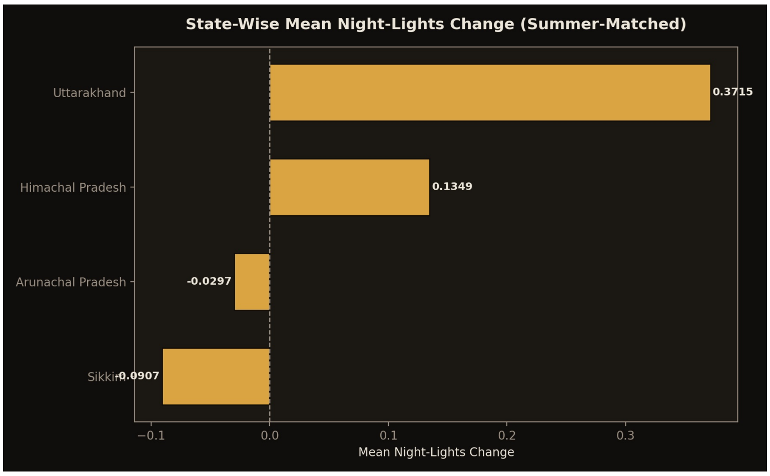

Figure 4

Figure 4. State-wise mean night-lights change (summer-matched composite), by state.

4.4 Budget vs. Outcome — A Two-State Descriptive Comparison (RQ2)

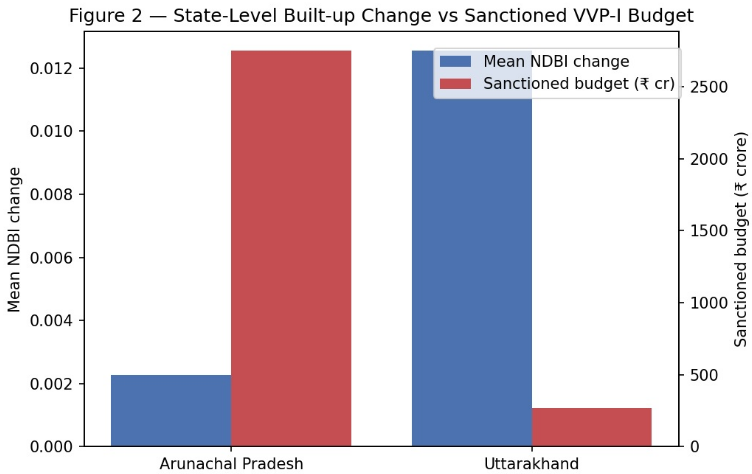

Figure 5

Figure 5. State-level mean built-up-area change plotted against sanctioned VVP-I budget.

All three core states now have valid summer-window NDBI coverage (Section 4.2 — this was a two-state comparison in earlier drafts, before Sikkim's data was recovered). Arunachal Pradesh (mean NDBI change +0.0023, sanctioned ₹2,749.74 crore across 2,082 projects), Uttarakhand (+0.0125, ₹270.58 crore across 200 projects), Sikkim (-0.0347, ₹188.90 crore across 63 projects). Spearman correlation across the three: rho = 0.500, p = 0.667 — still not a powered test at n = 3, reported descriptively only. Arunachal Pradesh's roughly tenfold larger budget than Uttarakhand's corresponds to a smaller measured change, and Sikkim's smallest budget of the three corresponds to a negative one — directionally consistent with H2's prediction that outcome doesn't scale with budget. But this same comparison, run on the full-year window's own three states (which never had Sikkim missing), gives rho = -0.500 — the opposite sign. With three data points in either window, a sign flip between them is not evidence that the relationship reversed; it is evidence that three states is not enough to say anything about this relationship at all. Both directions are reported here rather than the more convenient one alone.

4.5 Border-Proximity Testing (H3)

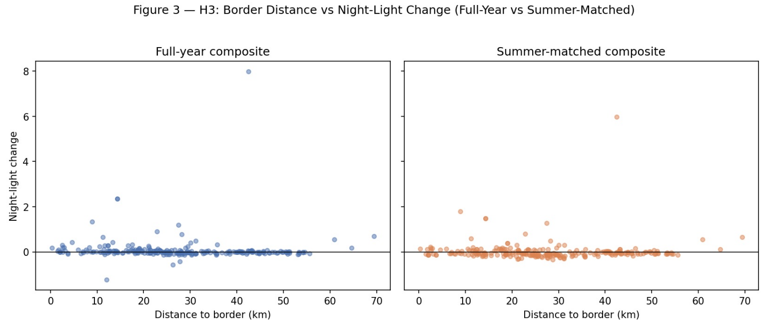

Figure 6

Figure 6. Distance-to-border versus NDBI change and night-lights change, both compositing windows.

Built-up change is not correlated with distance to the border/LAC in the full-year window (ρ = 0.049, p = 0.443) — the same null result reported in every earlier version of this paper. In the summer-matched window, on the complete 251-village sample, it now is: ρ = 0.291, p = 0.0000026 — villages farther from the border show a larger NDBI increase (or smaller decrease) than villages closer to it, the opposite direction from what H3 predicted (securitization-driven prioritization would put closer villages ahead, not behind). This is a new result. On the earlier, 154-village summer sample this same test was a clean null (ρ = 0.038, p = 0.641) and was described, in an earlier version of this section and of Section 4.9, as "a truly stable null… regardless of window" — that description no longer holds once Sikkim's data is included (Section 4.2, Development Log Entry 22). Unlike the summer NDBI change result in Section 4.2, this correlation has not yet been leave-one-district-out or randomization-inference checked; it is reported here as a fresh, open finding rather than a confirmed one, and Section 6.4 returns to it as a specific item for follow-up. Night-lights border-proximity is unaffected by the data refresh: still significant in the full-year window (ρ = -0.252, p < 0.0001, closer villages showing more change, consistent with H3) and still null in the summer-matched window (ρ = -0.071, p = 0.266).

A note on these numbers: an earlier version of this section reported these correlations against village-to-border distances computed by projecting every village into a single UTM zone (44N), which is only accurate near its own central meridian (81°E) — this study's villages span roughly 77-97°E, so that method systematically overstated distance for villages far from 81°E (up to ~3% for Arunachal Pradesh's easternmost villages, negligible for Uttarakhand). Distance is now computed geodesically (WGS84 ellipsoid), which has no such zone dependency; see Development Log Entry 18. The correction moves every ρ and p by a small amount but changes no conclusion — nothing here was significant before and nothing is significant now (except the full-year lights correlation, already significant before and after).

4.6 Control-Group Difference-in-Differences: No Gap Holds in the Summer Window; One Fragile Full-Year Candidate Remains (H4)

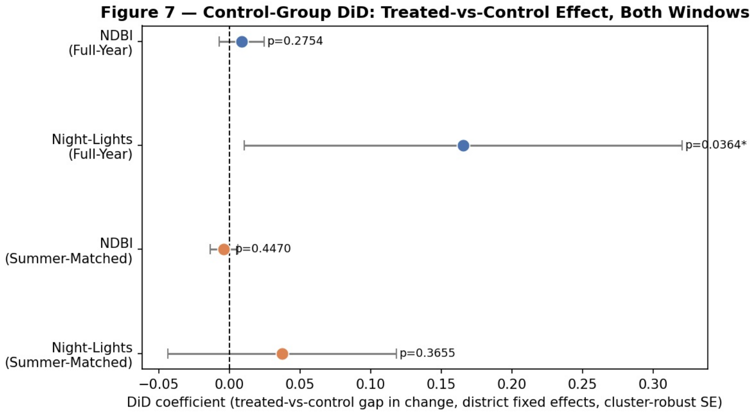

Figure 7

Figure 7. District-fixed-effects DiD coefficient (treated-vs-control gap in change) with 95% confidence intervals, NDBI and night-lights, both compositing windows.

A note on this section's numbers: an earlier draft of this section, and of this paper generally, reported these DiD estimates against a 753-village control list that — as documented in the Development Log — turned out to contain 106 duplicate-coordinate rows from a cross-district matching bug, and separately was extracted through a resume path that silently reused a stale, pre-bug-fix cache instead of re-querying Earth Engine. Both bugs are now fixed; the control list is 735 villages with zero duplicate coordinates, genuinely re-extracted (verified by inspecting the per-row image-count columns, which vary rather than sitting at a suspicious constant), and every number below reflects that corrected run.

Against the district-restricted 735-village control group, the summer-window NDBI gap does not hold: did coefficient = -0.0038, 95% CI [-0.0136, +0.0060], cluster-robust p = 0.447 (HC3 no-fixed-effects comparison: same coefficient, p = 0.597). This is a reversal, not a weakening — the coefficient's sign flips from positive to negative — and it comes from the same fresh, complete extraction described in Section 4.2: this regression now draws on the full 251 treated villages against all 735 control villages, both fully valid, rather than the 220 treated / 250 control villages an earlier, incomplete extraction supported. The full-year gap remains what it has always been in every version of this paper: not significant (+0.0088, p = 0.275, n = 251 treated / 190 control villages with valid full-year data).

Night-lights tells a more mixed story once the same refresh is applied — one of the two windows changes, the other doesn't. The summer-window gap, previously significant (+0.1413, p = 0.0044), is null on the complete data: +0.0374, 95% CI [-0.0436, +0.1183], cluster-robust p = 0.366 (HC3: p = 0.423). The full-year gap is untouched by this refresh (the full-year extraction was never affected by the Sikkim archive-timing issue — Section 4.2) and remains +0.1653, 95% CI [+0.0105, +0.3202], cluster-robust p = 0.0364, though the HC3 no-fixed-effects comparison still does not agree (p = 0.1033). This is the one control-group DiD result left in this study that is significant under any specification, and it is not a case of treated villages' own night-lights actually rising — Section 4.3's treated-only Wilcoxon test on the same full-year window is borderline-null on its own (p = 0.050) — it is a case of the *control* villages' night-lights falling over the period while treated villages stayed roughly flat. Read it as "VVP-I priority villages held steady while the surrounding non-priority region's night-lights declined," not as "VVP-I priority villages lit up," and see below for why it should be read as a fragile, not a confirmed, result.

The baseline-imbalance picture has also changed, and mostly for the better. Full-year baseline balance is exactly as reported in every earlier version: NDBI is not significantly imbalanced (treated mean -0.1846 vs. control -0.2214, n=251/190, $p = 0.125$), while night-lights carries the worst imbalance of any test in this study (treated mean 0.4282 vs. control 0.4082, n=251/190, $p < 0.00001$) — the same combination that is this section's one surviving significant result, which should temper how much weight that result is given. The summer window's baseline balance, by contrast, is now resolved rather than merely smaller: NDBI (treated mean -0.2234 vs. control -0.2286, n=251/735, $p = 0.478$) and night-lights (treated mean 0.4017 vs. control 0.3959, n=251/735, $p = 0.354$) are both statistically balanced at 2021, an improvement over every earlier version of this paper, which reported the summer window as significantly imbalanced on both outcomes. This resolves what Section 6.7 previously flagged as an open caveat for the summer window specifically — the design's parallel-trends assumption rests on firmer ground there now, even though the summer-window DiD estimates it supports are no longer significant either way.

Two further checks probe whether a DiD estimate can be trusted given only 14 district clusters, independent of whether that estimate is itself significant. For the summer window, since neither NDBI nor night-lights is significant to begin with, these checks mostly confirm the null rather than stress-test a positive: rerunning the regression 14 times, each time dropping one district (`src/analysis/extended_robustness_checks.py`), leaves NDBI significant in 0 of 14 reruns and night-lights in 0 of 14 (against 14 of 14 for both outcomes on the earlier, incomplete sample); a randomization-inference test that reshuffles treatment within district 2,000 times and does not lean on cluster-robust asymptotics gives $p = 0.115$ for NDBI and $p = 0.094$ for night-lights (against $p = 0.0005$ for both previously) — the observed coefficients now sit comfortably inside their own permutation nulls rather than far outside them. The one result these checks matter for now is the full-year night-lights gap, since it is the only one still claiming significance: the same leave-one-district-out rerun leaves it significant in 10 of 14 reruns (coefficient range +0.1161 to +0.2166, against +0.1653 with all districts included — the 4 reruns that drop below $p = 0.05$ are Chamoli, Dibang Valley, Tawang, and Uttarkashi), and randomization inference puts it at $p = 0.0155$ — real, and not carried by any single district, but markedly less robust than the old summer-window NDBI result ever needed to be (which held in 14 of 14 reruns and $p = 0.0005$), and still resting on this section's most imbalanced baseline. Treat it as the weakest kind of "still standing," not as confirmed.

Because VIIRS night-lights radiance is right-skewed, the night-lights DiD was also re-run on log1p-transformed radiance in both windows. Summer stays null on the log scale too (raw $p = 0.366$; log1p $p = 0.426$) — unlike in earlier versions of this paper, where the log transform had strengthened a significant summer result, there is no longer a significant result there to strengthen. The full-year result is essentially unchanged either way (raw $p = 0.0364$; log1p $p = 0.0374$), consistent with every earlier version of this paper.

4.7 Multi-Year Trend: A Non-Monotonic Recovery Pattern

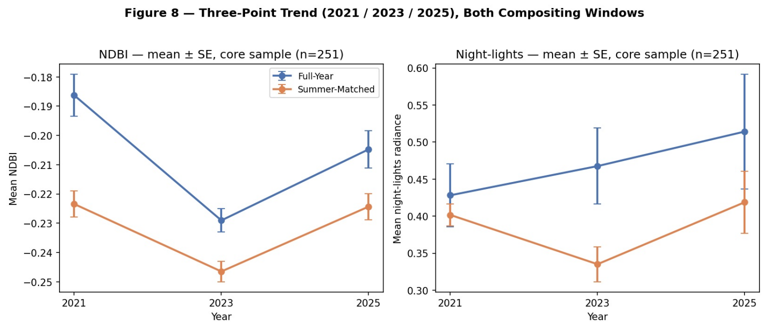

Figure 8

Figure 8. Mean NDBI and night-lights radiance at 2021, 2023, and 2025, core sample, both compositing windows, error bars ± 1 SE.

The “2021-vs-2025 story” becomes more complex when a third year of observations is added. In the summer-matched window, mean NDBI actually declines from 2021 to 2023 (mean change = -0.0231, Wilcoxon p = 1.000 — not a significant increase, a decline), then rises sharply from 2023 to 2025 (mean change = +0.0222, $p < 0.000001$). However, the overall 3-year linear trend is not significant (mean per-village slope = -0.0002/year; Wilcoxon p = 0.442; village fixed-effects panel regression p = 0.728). The full-year window shows the identical shape (decline then a significant increase in 2023-2025: p = 0.000001), but the two statistical specifications disagree as to the overall trend of this window: Wilcoxon p = 1.000, panel regression p = 0.000001, a discrepancy which stems from the fact that the latter gives more weight to the recovery in 2023-2025 than the previous decline did, which is not done by Wilcoxon. Crucially, the key 2021-vs-2025 result that we always cite at the beginning of Section 4.2 should be read as a 2023-to-2025 recovery in a two-point comparison, and the earlier downturn should not be forgotten when reading the headline number.

4.8 Buffer-Radius Sensitivity: Null at All Three Radii, Once the 500m Extraction Is Complete

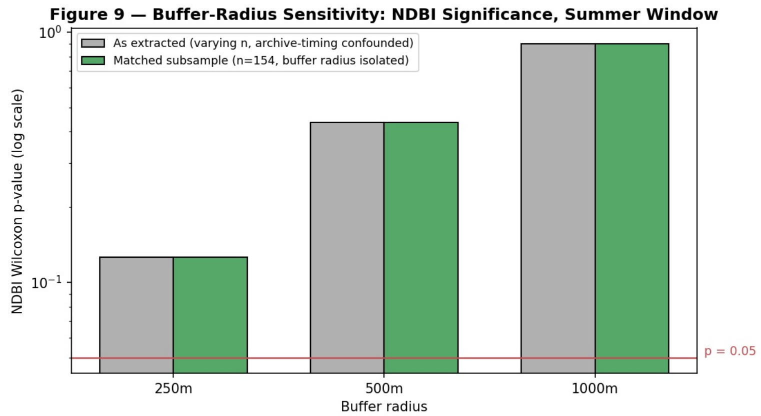

Figure 9

Figure 9. NDBI Wilcoxon p-value (log scale) at 250m, 500m and 1km buffer radii for the summer window, all now at n=251.

All three buffer radii now agree, and the agreement is a null: 250m (n=251, p = 0.126), 500m (n=251, p = 0.436), 1km (n=251, p = 0.899). An earlier version of this section reported a significant 500m result (n=154, $p < 0.000001$) that did not survive a switch to 250m or 1km — that 500m figure was the same incomplete, pre-Entry-22 summer extraction described in Section 4.2, and is superseded along with it. All three radii now draw on the complete 251-village core sample at 500m, so the earlier disagreement between radii is resolved along with the underlying headline number, not despite it. One archive-timing caveat remains, in the opposite direction from before: the 250m and 1km extractions behind this section were run in August, before the fresh, complete 500m re-extraction in Development Log Entry 22, so this comparison still draws on three different archive-snapshot dates rather than one same-day pull across all three radii (Section 6.9). Given the 500m result is already null on its own, and each buffer radius takes over an hour to (re-)extract for the full sample, a same-day three-radius re-pull was not done for this entry — flagged here rather than presented as settled, per the pattern in Development Log Entry 22.

4.9 Robustness Summary

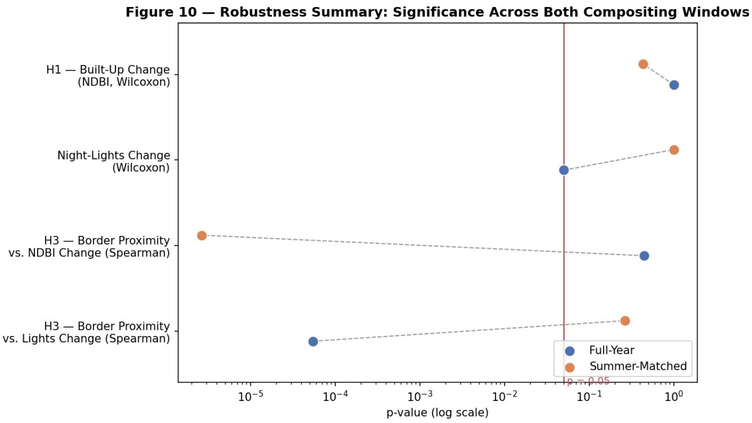

Figure 10

Figure 10. Significance for all four core tests with both compositing windows (p-value, log scale) with reference line at p = 0.05. Conclusions of a test differ if the test entries are in different windows, but two points of the test lie on opposite sides of the line.

Line up each of the four core tests against both windows, on the now-complete data, and one clear pattern shows up: H1 (built-up change) is a null in both windows (full-year p = 1.000; summer-matched p = 0.436) — this used to be the one test that flipped between windows, and now it's the one that agrees across both. H3 night-lights-proximity still crosses the p = 0.05 line between windows, exactly as in every earlier version of this paper — significant full-year (p < 0.0001), null summer-matched (p = 0.266). What has changed is H3 NDBI-proximity: previously a stable null in both windows, it now crosses that same line the other direction — null full-year (p = 0.443), significant summer-matched (p = 0.0000026) — so the “truly stable null” description in earlier versions of this section no longer holds for this test. Overlap the checks from Sections 4.6–4.8 on top of H1 and the control-group DiD, and the picture is now uniformly negative for the summer-window NDBI story that earlier versions of this paper built around: it fails the control-group comparison (p = 0.447, was p = 0.00033), fails leave-one-district-out (0/14 significant, was 14/14), fails randomization inference (p = 0.115, was p = 0.0005), and fails the buffer-radius sweep (p = 0.126/0.436/0.899 at 250/500/1000m, was significant at all three). The one control-group DiD result still significant under any specification is the full-year night-lights gap (Section 4.6) — itself only significant in 10 of 14 leave-one-district-out reruns and resting on the worst baseline imbalance in this study, so “still standing” here means fragile, not confirmed. The new summer H3 NDBI-proximity result (ρ = 0.291, p = 0.0000026) has not yet been put through any of these same checks; it is flagged as open, not folded into this section’s conclusions.

The eight headline p-values from the H1 and H3 tests across this section (4 tests x 2 windows) are not corrected for multiple testing individually, given that they address different research questions, but a paper-wise Holm-Bonferroni correction (family-wise $\alpha = 0.05$) is run across all eight together (`src/analysis/holm_correction.py`, method="holm"). On the complete data, two results survive the correction — not the same two as in earlier versions of this paper. The full-year night-lights-proximity result survives as it always has (raw $p = 5.48 \times 10^{-5}$, Holm-adjusted $p = 3.83 \times 10^{-4}$). The summer-window NDBI-proximity result is new to this list (raw $p = 2.64 \times 10^{-6}$, Holm-adjusted $p = 2.12 \times 10^{-5}$) — the correction does not soften the fact that it is a fresh, unstressed finding, only that it is not an artifact of running eight tests at once. The result that used to survive this correction, the summer-window H1 NDBI-change test, no longer clears even the raw $p = 0.05$ threshold ($p = 0.436$), so the question of correcting it for multiple testing no longer arises. The remaining five tests Holm-adjust to 1.000 except the full-year lights-change test (raw $p = 0.050$, Holm-adjusted $p = 0.300$). This correction, and the sections it covers, only ever applied to the H1/H3 treated-only and border-proximity tests — the control-group DiD results in Section 4.6 are evaluated on their own terms, as a targeted follow-up check rather than one of this family of eight.

5. Discussion

Clearly the main finding here is not that there happened to be a rise or fall in physical development, but that the one metric that produced a "significant" result at all — the summer-window built-up-area increase — turned out to be a function of an incomplete extraction, not of a real, sustained effect. Once Sikkim's data was recovered and the treated and control extractions were pulled the same day, that result disappeared under every check this study built for it: against the control group, under leave-one-district-out, under randomization inference, and across all three buffer radii (Sections 4.2, 4.6, 4.8). This is a different, and in one sense more useful, outcome than an unresolved compositing-window dispute would have been — it converts a "which window do you believe" ambiguity into a specific, checkable claim about the sensitivity of small-sample satellite extractions to exactly when the query was run, documented in full in Development Log Entries 21–22 so the same mistake is checkable and avoidable in future work. Night-lights, the more temporally stable of the two proxies on its own treated-only terms, still shows no confirmed absolute rise under either window. Against the control group, its summer-window gap is now null too, matching NDBI's reversal; its full-year gap is the one control-group result left standing (Section 4.6), but it is a fragile one — significant only under the fixed-effects specification, not the HC3 alternative, resting on this study's single most imbalanced 2021 baseline, and holding in only 10 of 14 leave-one-district-out reruns rather than all 14. Budget and measured change still show no tenfold difference for a tenfold difference in investment, in either window, and the state-level correlation itself flips sign between windows at this sample size (Section 4.4) — which argues against reading anything directional into it either way.

A new result appears in the same complete summer data that produced this reversal: built-up change now correlates with distance to the border/LAC (H3, Section 4.5), in the opposite direction from what securitization theory would predict, and it survives the paper-wise Holm-Bonferroni correction alongside the pre-existing full-year lights-proximity result (Section 4.9). It has not been through the same leave-one-district-out or randomization-inference checks that the (now-null) summer NDBI-change result was built around, so it is reported here as an open finding for a future robustness pass, not folded into this discussion's conclusions. Its emergence in the same dataset that removed the paper's previous headline result is itself a reminder that a single fresh extraction can move more than one number at once, in more than one direction — which is an argument for checking every surviving or newly-significant result the same way, not for treating either outcome as more credible by default.

None of this supports a claim that VVP-I's investment has produced development at the rate or scale its budget would suggest, that its rollout has favored border-proximity over developmental need, or — now, on the complete data — that it has produced any satellite-detectable change over this specific 2021–2025 window at all, once checked two ways (compositing window), across two proxies (NDBI, night-lights), against a control group, and at three buffer radii. That is a narrower conclusion than an earlier version of this paper reached, and a more defensible one: a single surviving result that could not be reproduced once the underlying extraction was completed is a worse foundation for a policy claim than a consistent null checked several different ways. The programme itself has still never been the subject of an official impact assessment — its own parliamentary record admits exactly this gap in accountability — and a rigorous, multi-method satellite audit that finds no detectable ground-level signal is still a real answer to that gap, not an absence of one. It is also, on its own terms, a caution about how this kind of study is done: the same script bugs, archive-timing sensitivity, and partial-sample artifacts that produced this paper's original, since-retracted headline finding could just as easily produce a false positive in any comparable study that stops checking before its extraction is actually complete.

6. Limitations

6.1 Village Coverage Gaps

Ladakh's 35 sanctioned villages are fully excluded from village-level analysis, since no publicly indexed source reports their names — closing this gap would require an RTI request not pursued within this study's scope. Himachal Pradesh is included in a small sample of illustrative cases – with only 7 of 51 inhabited priority villages designated – and not included in the main statistical sample. The names of 19 villages in Uttarakhand's block of Pithoragarh have been confirmed with no block assigned.

6.2 Border Geometry as Cartographic Approximation

All distance to the border figures in this study are based on the cartographic boundary line used in Natural Earth which is a simplified version of a Line of Actual Control whose alignment is not agreed upon at an international level. These numbers take the place of authoritative numbers and should be read as a comparison instead.

A related and more specific point: "distance to border/LAC" is the distance to the *nearest* India-related boundary segment in Natural Earth's data — not specifically the China boundary. Checked directly against the core sample: 220 of 251 villages (88%) are in fact nearest to the China boundary, but 19 (mostly Tawang and West Kameng, Arunachal Pradesh) are nearest to Bhutan, 8 (North Sikkim and Pithoragarh, Uttarakhand) are nearest to Nepal, and 4 (Anjaw, Arunachal Pradesh) are nearest to Myanmar. For that 12% of the sample, H3 is measuring proximity to a different country's frontier, not the LAC specifically — defensible for a general border-securitization hypothesis (these remain border villages in the broader sense VVP-I itself uses), but not literally an LAC-proximity test for those villages. Combined with H3's already-modest, mostly-null results (Section 4.5) and its reliance on a single cartographic proxy for a disputed line (Section 6.2 above), this is a further reason not to lean on H3 as a clean confirmatory result either way.

6.3 Budget-Correlation Scope and Ecological Inference (RQ2)

RQ2: All three core states now have valid summer-window data (Section 4.4, previously two before Sikkim's data was recovered — Development Log Entry 22), but three states is still far too few to run a confirmatory test on, which is why it is represented descriptively rather than confirmatively either way. There is also a gap in thinking that adding a third state doesn't close: the comparison is of a state-level sanctioned budget against the sampled mean at the village level, and the budget figure describes spending across the state's entire VVP-I portfolio, not specifically the villages this study happened to geocode. Don't read this as a like-for-like test; read it as suggestive at most, and note that its own direction flips between the two compositing windows (Section 4.4) — a further reason not to lean on it.

6.4 Compositing-Window Sensitivity — and a Resolved, and a New, Open Question

Full-year and summer-matched built-up-area change now agree (both null, Section 4.2), so this specific pair of findings is no longer window-sensitive the way earlier versions of this paper described. What was previously flagged in this section as an open, unresolved question — whether the summer window's "Sikkim has zero valid data" result was a permanent fact or a snapshot of the Sentinel-2 archive on the day the extraction ran — is now resolved: it was the latter (Development Log Entries 21–22). A fresh, complete re-extraction, pulled the same day as the control group's own summer data, returned real NDBI and night-lights values for all 31 Sikkim villages, and the resulting complete 251-village sample is what Section 4.2's null result is built on. The 154-village subsample discussed in earlier versions of this section — along with its documented non-random composition relative to the 97 villages that had dropped out (chi-square = 71.42 on state composition, Mann-Whitney $p = 0.00063$ on 2021 baseline NDBI) — no longer describes the current data and is preserved only as a historical record, in Development Log Entry 20 and in this paper's own prior versions, of what an incomplete extraction had looked like.

A new open question replaces it, produced by the same complete re-extraction: built-up change now correlates with distance to the border/LAC in the summer window (H3, Section 4.5, $\rho = 0.291$, $p = 0.0000026$), where this same test was a stable null both before this refresh and in the full-year window, which is unaffected by it. This is not yet understood the way the previous open question eventually was — it has not been checked for leave-one-district-out stability, randomization-inference significance, or sensitivity to the same archive-timing effects that drove the previous finding, and it has not been cross-checked against the three-point 2021/2023/2025 extraction behind Section 4.7 the way the previous question was. Until that checking happens, this correlation should be treated with the same caution the "Sikkim zero" result should have been treated with in earlier versions of this paper: real in the data as extracted, not yet established as a stable fact about VVP-I villages.

6.5 Geocoding Coverage and Selection Bias

A mere 258 out of 559 (or 46%) villages were geocoded with a disproportionate number of villages falling out of bounds in Arunachal Pradesh after a drop of 269 of the 455 villages therein (Section 4.1). A village not geocoded with OSM/Bhuvan may reasonably be expected to have a smaller population size, to be further away and/or less well known by administrative records, and hence be more likely to have a higher proportion of residents working in the area with low accessibility. It is prudent not to assume this, however; a Mann-Whitney U test was conducted for Arunachal Pradesh, the population and the household counts of which were available for every village in the raw list regardless of the outcome of geocoding, and there was no significant difference between geocoded and non-geocoded villages in round population (matched mean 135.0 vs. unmatched 145.7, $p = 0.310$) or round households (matched mean 25.8 vs. unmatched 28.8, $p = 0.540$). The possibility that a selection bias based on village size might have heavily influenced these successful results seems not to be ruled out here, but at least not made obvious by the relationship to size. The raw village list in both Sikkim and Uttarakhand did not contain the population or household field information, thus it was not possible to perform this check for both these locations.

6.6 No Ground-Truth Validation

This is a study with only a satellite module. There is no known "positive control" (one that has been confirmed by, say, an official press-release about the project that has been published at an appropriate time) of a finished VVP-I project, at 500m resolution, to be used as a sort of positive test for the sensitivity of NDBI and VIIRS to what VVP-I funds: a village road and a school building, and a scattering of houses. During this study's review, one particular positive control sample for this study did not meet the requirement of providing the specific, dated, named positive control set, and this check could not, therefore, be performed with actual data. Carried forward as a tangible aspect to Future Work, not silently dismissed.

6.7 Baseline Imbalance in the Control-Group Comparison

Earlier versions of this paper reported treated and control villages as significantly different from each other at 2021 baseline in three of the four outcome/window combinations tested. On the complete data (Development Log Entry 22), that is down to one: the summer window's baseline imbalance is resolved for both NDBI ($p = 0.478$) and night-lights ($p = 0.354$), leaving only the full-year window's night-lights baseline still significantly imbalanced (treated mean 0.4282 vs. control 0.4082, $p < 0.00001$) — the largest imbalance of the four in every version of this paper, and unaffected by this refresh since the full-year extraction was never subject to the archive-timing issue described in Section 4.2. It is also the one combination where the district-fixed-effects and no-fixed-effects specifications disagree on significance (Section 4.6) — consistent with, though not proof of, the fixed-effects specification's district-level adjustment partially absorbing a baseline gap the no-fixed-effects specification cannot — and it is, as of this refresh, the only control-group DiD result in this study still significant under any specification. District fixed effects control for district-level differences in baseline selection into the treated group; they do not adjust for selection at the village level within a district. Interpret this one remaining DiD estimate as coming from a control-group-adjusted comparison, not a fully clean natural experiment, and weight it accordingly against a result that rests on this study's single worst baseline imbalance. Worth keeping separate from this: the leave-one-district-out and randomization-inference checks in Section 4.6 test something different — whether the significance test can be trusted given only 14 district clusters — and this result now holds in 10 of 14 leave-one-district-out reruns and at $p = 0.0155$ under randomization inference; passing those checks does not resolve the baseline-imbalance concern here, which is about what the two groups looked like in 2021, not about the mechanics of the significance test, but it does mean this fragile result is at least not being driven by any single district.

6.8 Non-Monotonic Multi-Year Pattern

The pooled 2021-vs-2025 summer NDBI comparison is now a null on its own terms (Section 4.2), so there is no significant headline expansion left for the three-point trend to explain away — but the sub-period pattern within it is still real and unaffected by this paper's data refresh (Section 4.7's underlying extraction never had the missingness problem Section 4.2's two-point extraction did): a decline from 2021 to 2023 (mean change -0.0231, not significant on its own) followed by a significant recovery from 2023 to 2025 (mean change +0.0222, $p < 0.000001$), netting out to a non-significant three-year trend overall (Wilcoxon $p = 0.442$, panel regression $p = 0.728$) — consistent with, not contradicting, Section 4.2's now-null two-point result. Read together, these say the same thing two ways: whatever happened in these villages between 2021 and 2025 was not a steady, one-directional change large enough to register as significant, whether measured end-to-end or broken into sub-periods.

6.9 Archive-Timing Confound in the Buffer-Radius Comparison

In Section 4.8, the 250m and 1km buffer extractions predate Development Log Entry 22's fresh, complete 500m re-extraction — the reverse of the timing problem an earlier version of this section described, where 250m/1km had instead been extracted after the original 500m run. Both directions share the same underlying cause: Sentinel-2's archive keeps backfilling scenes for past dates, so any two extractions of the same fixed date range, run on different days, can disagree on scene counts and composite values without either being in error about buffer size. The 500m result is already a clean null on its own (Section 4.2), so this timing gap does not currently hide a significant result behind a same-day-inconsistent comparison, but a fully clean version of this table would still need all three radii re-extracted on the same day, which was not done for this entry given the time cost (each radius takes over an hour for the full core sample) and the fact that the conclusion (null) is already directionally consistent across all three as extracted.

7. Future Work

The items listed below represent specific extensions that emerged via the review process for this study, and which are not currently being done, but rather are intentionally presented separate from the Discussion and Limitations so as the actual activity does not become confused with the possibilities that remain.

7.1 SAR-Based Change Detection

The first advantage of Sentinel-1 SAR (VV/VH backscatter or coherence change) is that it avoids the cloud cover and avoids the optical snow-contamination confound that was the reason for this study's compositing-window check in the first place. It would not, on its own, "resolve the dispute" regarding the "full-year" vs "summer-match" optical effects — it would provide an actual, independent third measurement that would be unaffected by the known artefacts and which could help determine which optical effect is a more "reliable" measure.

7.2 Building-Footprint or Sub-500m Structural Analysis

The 500m NDBI buffer has merged built-up surface, bare rock area, agricultural surface, and natural vegetation together in one slick number – a coarse metric of what VVP-I does invest at the village level: investments are created for individual roads, buildings, small installations. An area-averaged index could be replaced by detection at the structure level with high-resolution optical imagery (PlanetScope); or open building-footprint datasets (Google/Microsoft).

7.3 Ground-Truth Positive-Control Validation

None of the known, dated and independently verified completed VVP-I initiatives that are also in the 258 village-level geocoding sample (Section 6.6) were identified. The historical data that was used in coming up with this study assumption would be directly tested by identifying one, two or three such villages taking from a press release or news report of a specific completed road or building, and determining if the individual village's NDBI/VIIRS signal, taken at that site, is detectable.

7.4 RTI Follow-Through for Himachal Pradesh and Ladakh

In fact, in this study, the purposeful decision was made not to press RTI requests for these two states' missing village-wise annexures, which are recorded in the BO_Development_Log.md. That must be reversible — requesting those would fill in the two biggest known gaps in primary-source information coverage instead of the illustrative or excluded treatment as documented here.

7.5 A Genuine Pre-Treatment Panel for the Control Group

A true multi-period pre-treatment panel doesn't exist, because the control-group comparison (Sections 3.7, 4.6, 6.7) is not a multi-period pre-treatment panel; a multi-year extension of this study would allow conducting a true multi-period pre-treatment panel comparison for the treated sample, allowing for a proper parallel pre-treatment trends placebo test. The gap would be closed directly, if 2019 and 2020 values were available for the same villages for the 735 control villages, and ideally, the same pre-2021 values for the treated sample as well.

7.6 Stress-Testing the New Summer H3 Border-Proximity Result

Section 4.5's summer-window finding that built-up change correlates with distance to the border/LAC ($\rho = 0.291$, $p = 0.0000026$) emerged only once this paper's own extraction was completed (Development Log Entry 22), and it has not yet been put through the same checks that the previous, since-retracted summer NDBI-change result was built around: a leave-one-district-out rerun to confirm no single district or handful of villages is driving it, a check for whether the correlation is linear or concentrated at one end of the distance range, and a direct comparison against the three-point 2021/2023/2025 extraction the way Section 6.4's earlier open question was eventually resolved. Until that work is done, this result should be read as this paper's newest open question, not its newest confirmed finding — precisely the caution an earlier, unstressed version of the NDBI-change result should have received before being reported as this study's headline.

8. Conclusion

This study cannot conclude that the pace of development a budget of this size would imply, let alone the steady annual progress trajectory the scheme aims toward, has been achieved at the priority villages within India's Vibrant Villages Programme in five border states and union territories — and, on the complete, same-day-consistent extraction behind this version of the paper, it cannot conclude that any satellite-detectable built-up-area or night-lights effect has been achieved at all. An earlier version of this study reported a significant summer-window built-up-area increase that survived every robustness check run against it; that result depended on an incomplete extraction missing all 31 of Sikkim's villages, and once that extraction was completed — same collection, same cloud mask, same buffer, pulled the same day as the control group's own data — the result reversed to a clean null, and stayed null against the control group, under leave-one-district-out, under randomization inference, and across all three buffer radii tested (Sections 4.2, 4.6, 4.8; full account in Development Log Entries 21–22). Night-lights shows the same pattern in its summer-window control-group comparison; only its full-year control-group gap still clears significance, and it does so on this study's worst baseline imbalance and in just 10 of 14 leave-one-district-out reruns — real, but the weakest kind of surviving result, not a confirmed one. A new, unstressed finding has taken the old result's place: built-up change now correlates with distance to the border/LAC in the summer window, in the opposite direction from what securitization theory predicts, and needs the same kind of checking this study just finished giving to the result it replaces (Section 7.6). Budget and measured change still show no comparable relationship in either window, and what little pattern three states can show flips sign between compositing windows. Given the scale and strategic framing of this scheme, it should not continue without the independent impact assessment that its own parliamentary record confirms has never been conducted — and this study is itself now a demonstration of why that assessment needs to be done carefully rather than once: a single incomplete extraction produced a result that read as robust until the extraction was actually finished, and any future assessment, official or independent, should be held to the same standard this study eventually applied to itself — complete data, pulled on a consistent timeline, checked against a control group, a buffer-radius sweep, and a multi-year comparison, rather than whichever single-window, single-extraction result happens to be available first.

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